

MarginScout Knowledge Base

Corn and soybean cost benchmarks and fertilizer reference data

This document is a structured reference for the MarginScout farm financial planning platform. It contains regional cost-of-production benchmarks for corn, historical fertilizer prices, and state-level fertilizer application rates for corn and soybeans. All data is from USDA Economic Research Service (ERS) and USDA National Agricultural Statistics Service (NASS).

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Sources: USDA, Economic Research Service — Commodity Costs and Returns data product. USDA, National Agricultural Statistics Service — Agricultural Chemical Usage and Agricultural Prices. Last updated: May 2026.

Section 1 — Corn cost and return benchmarks by region, 2025

All values are dollars per planted acre unless otherwise indicated. The seven regions are USDA-ERS Farm Resource Regions. The Heartland region corresponds to the central Corn Belt (Iowa, Illinois, Indiana, Ohio, Missouri).

Line item	Heartland	N. Crescent	N. Great Plains	Prairie Gateway	Eastern Uplands	Southern Seaboard	U.S. total
Gross value — grain	\$837.39	\$703.54	\$770.43	\$671.00	\$757.13	\$855.67	\$779.80
Gross value — total	\$839.24	\$711.55	\$774.63	\$672.40	\$760.84	\$856.23	\$782.44
Seed	\$119.55	\$109.66	\$104.41	\$92.76	\$127.65	\$103.20	\$111.11
Fertilizer	\$169.71	\$180.77	\$144.50	\$114.49	\$187.64	\$213.83	\$158.88
Chemicals	\$49.23	\$44.96	\$42.66	\$55.12	\$47.74	\$36.97	\$48.86
Custom services	\$37.95	\$41.99	\$39.08	\$33.25	\$16.44	\$36.41	\$37.01
Fuel, lube, electricity	\$25.59	\$30.13	\$19.66	\$30.10	\$23.34	\$46.28	\$26.71
Repairs	\$47.04	\$48.46	\$44.98	\$54.78	\$46.18	\$51.13	\$48.70
Interest on op. capital	\$8.89	\$9.03	\$7.83	\$7.56	\$8.89	\$9.66	\$8.54
Total operating costs	\$457.96	\$465.00	\$403.23	\$389.55	\$457.89	\$497.76	\$440.11
Hired labor	\$3.50	\$6.24	\$3.74	\$6.99	\$2.27	\$9.39	\$4.66
Unpaid labor opp. cost	\$32.11	\$36.33	\$26.44	\$24.80	\$48.47	\$69.82	\$31.37
Machinery cap. recovery	\$173.63	\$162.71	\$167.61	\$175.01	\$162.77	\$158.68	\$171.94
Land opp. cost	\$229.57	\$154.90	\$135.17	\$141.39	\$137.35	\$104.70	\$190.30
Taxes & insurance	\$21.86	\$20.81	\$15.67	\$21.15	\$24.06	\$17.77	\$20.72
General farm overhead	\$28.66	\$33.94	\$20.93	\$19.30	\$24.77	\$25.50	\$26.33
Total allocated overhead	\$489.33	\$414.93	\$369.57	\$388.64	\$399.70	\$385.86	\$445.32
TOTAL COSTS	\$947.29	\$879.93	\$772.80	\$778.20	\$857.59	\$883.62	\$885.43
Net of operating costs	\$381.28	\$246.55	\$371.40	\$282.85	\$302.95	\$358.47	\$342.33
NET VALUE	\$-108.05	\$-168.38	\$1.83	\$-105.80	\$-96.74	\$-27.39	\$-102.99
Yield (bu/ac)	214	178	208	168	176	173	198
Price (\$/bu)	\$3.92	\$3.95	\$3.71	\$4.00	\$4.30	\$4.95	\$3.93
Farm size (ac)	274	184	599	499	105	255	296

Notes: Fertilizer cost includes commercial fertilizers, soil conditioners, and manure. Custom services cost includes custom operations, technical services, and commercial drying. Net value shows farm profitability before government payments and after all economic costs including opportunity costs of land and unpaid labor. A negative net value indicates total economic costs exceeded gross value of

production.

Section 2 — Corn cost trends, Heartland region, 2016–2025

The Heartland region (Iowa, Illinois, Indiana, Ohio, Missouri) accounts for the majority of U.S. corn production. This table shows how key cost components have evolved over the past decade. All values are dollars per planted acre.

Line item	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Seed	\$104.52	\$103.26	\$99.46	\$97.18	\$97.27	\$104.82	\$117.77	\$123.92	\$119.61	\$119.55
Fertilizer	\$131.49	\$115.61	\$118.85	\$128.67	\$115.46	\$127.12	\$239.38	\$190.07	\$162.46	\$169.71
Chemicals	\$36.56	\$35.05	\$34.53	\$33.88	\$31.97	\$41.74	\$63.70	\$61.87	\$49.34	\$49.23
Fuel/lube/elec	\$20.40	\$23.21	\$26.28	\$25.29	\$21.40	\$22.77	\$32.85	\$28.35	\$26.53	\$25.59
Repairs	\$30.33	\$30.89	\$31.97	\$32.97	\$33.44	\$39.45	\$43.90	\$45.53	\$46.05	\$47.04
TOTAL OP. COSTS	\$347.34	\$332.35	\$337.60	\$345.00	\$323.62	\$366.08	\$537.95	\$497.67	\$451.49	\$457.96
Machinery cap. recovery	\$116.55	\$118.97	\$126.37	\$125.01	\$126.35	\$147.71	\$154.84	\$165.09	\$172.46	\$173.63
Land opp. cost	\$193.15	\$190.98	\$192.88	\$193.03	\$192.31	\$198.86	\$214.04	\$224.71	\$231.77	\$229.57
TOTAL ALLOC. OVERHEAD	\$364.76	\$366.48	\$378.29	\$379.79	\$382.39	\$417.51	\$446.53	\$470.98	\$488.02	\$489.33
TOTAL COSTS	\$712.10	\$698.83	\$715.89	\$724.79	\$706.01	\$783.59	\$984.48	\$968.66	\$939.51	\$947.29
NET VALUE	\$-65.05	\$-51.91	\$-40.28	\$-9.38	\$-22.88	\$226.59	\$318.02	\$25.60	\$-115.26	\$-108.05
Yield (bu/ac)	197	200	199	188	190	201	202	203	208	214
Price (\$/bu)	\$3.28	\$3.23	\$3.39	\$3.80	\$3.59	\$5.02	\$6.42	\$4.88	\$3.95	\$3.92

Key insight: Heartland total cost per planted acre rose from \$678 in 2016 to \$947 in 2025 (+40% over the decade). Fertilizer was the most volatile line item, ranging from \$111 (2017) to \$226 (2022) in the broader U.S. data. Net value has been negative in 7 of the last 10 years, reflecting the structural margin compression that motivates this product.

Section 3 — National corn yield, price, and net value history, 1996–2025

U.S. total values, useful for downside-scenario calibration: the historical range of corn prices, yields, and net returns sets defensible bounds for what-if modeling. All cost/value figures are dollars per planted acre; yield is bushels per planted acre; price is dollars per bushel at harvest.

Year	Yield (bu/ac)	Price (\$/bu)	Gross value	Operating costs	Total costs	Net of op. costs	NET VALUE
2025	198	\$3.93	\$782.44	\$440.11	\$885.43	\$342.33	\$-102.99
2024	190	\$3.97	\$757.97	\$434.66	\$878.31	\$323.31	\$-120.34
2023	186	\$4.92	\$920.41	\$479.74	\$908.81	\$440.67	\$11.60
2022	180	\$6.48	\$1,170.19	\$519.11	\$927.86	\$651.07	\$242.33
2021	186	\$5.07	\$944.56	\$353.33	\$735.19	\$591.24	\$209.37
2020	178	\$3.61	\$644.71	\$319.44	\$678.28	\$325.27	\$-33.57
2019	173	\$3.83	\$664.94	\$339.39	\$692.23	\$325.55	\$-27.29
2018	183	\$3.42	\$628.11	\$332.73	\$685.11	\$295.38	\$-57.00
2017	184	\$3.25	\$600.03	\$326.26	\$666.28	\$273.77	\$-66.25
2016	183	\$3.29	\$603.92	\$340.55	\$678.48	\$263.37	\$-74.56
2015	167	\$3.66	\$612.60	\$333.80	\$675.33	\$278.80	\$-62.73
2014	170	\$3.54	\$603.18	\$356.92	\$689.80	\$246.26	\$-86.62
2013	156	\$4.61	\$720.51	\$355.60	\$676.45	\$364.91	\$44.06
2012	118	\$6.79	\$802.55	\$349.59	\$653.57	\$452.96	\$148.98
2011	146	\$5.73	\$837.77	\$332.33	\$613.46	\$505.44	\$224.31
2010	159	\$4.33	\$689.39	\$286.41	\$550.20	\$402.98	\$139.19
2009	156	\$3.59	\$561.22	\$295.01	\$550.70	\$266.21	\$10.52
2008	144	\$4.36	\$629.36	\$295.69	\$529.38	\$333.67	\$99.98
2007	143	\$3.27	\$468.94	\$228.99	\$443.97	\$239.95	\$24.97
2006	138	\$2.54	\$351.87	\$205.98	\$409.74	\$145.89	\$-57.87
2005	149	\$1.74	\$260.43	\$186.37	\$386.88	\$74.06	\$-126.45
2004	169	\$2.13	\$362.35	\$175.94	\$377.50	\$186.41	\$-15.15
2003	149	\$2.13	\$319.62	\$161.16	\$354.41	\$158.46	\$-34.79
2002	134	\$2.32	\$312.82	\$145.48	\$334.31	\$167.34	\$-21.49
2001	144	\$1.84	\$266.92	\$162.30	\$348.53	\$104.62	\$-81.61
2000	138	\$1.77	\$246.67	\$164.99	\$378.32	\$81.68	\$-131.65
1999	135	\$1.69	\$230.70	\$156.92	\$364.73	\$73.78	\$-134.03
1998	136	\$1.91	\$262.88	\$157.71	\$362.86	\$105.17	\$-99.98

Year	Yield (bu/ac)	Price (\$/bu)	Gross value	Operating costs	Total costs	Net of op. costs	NET VALUE
1997	130	\$2.52	\$331.37	\$162.25	\$363.73	\$169.12	\$-32.36
1996	130	\$2.82	\$369.93	\$160.95	\$353.94	\$208.98	\$15.99

Downside scenario calibration. Over the 30-year history: yield ranged 130–198 bu/ac (median ~155), price ranged \$1.69–\$6.79/bu (median ~\$3.55), and net value ranged -\$134 to +\$242 per acre. A defensible worst-case scenario for a present-day farmer is 20% price decline from current + 10% yield decline, both within historical bounds.

Section 4 — Average U.S. farm prices of selected fertilizers

Historical NASS-reported farm-level fertilizer prices (\$/material short ton). The NASS price-paid series ended in April 2014. For post-2014 prices, refer to USDA AMS state-level Production Cost Reports (bi-weekly publication). The series below establishes the long-term context for current price benchmarking.

Year	Month	Anhydrous ammonia (30%)	Urea (44-46% N)	Ammonium nitrate	Sulfate of ammonia	Super-phos. (20%)	Super-phos. (44-46% P)	DAP (18-46-0)	Potassium chloride (60% K)	
1990	Apr.	\$199.00	\$132.00	\$184.00	\$180.00	\$154.00	—	\$201.00	\$219.00	\$155.00
1992	Apr.	\$208.00	\$141.00	\$198.00	\$178.00	\$151.00	—	\$206.00	\$224.00	\$150.00
1994	Apr.	\$243.00	\$137.00	\$207.00	\$196.00	\$170.00	—	\$212.00	\$224.00	\$146.00
1996	Apr.	\$303.00	\$182.00	\$278.00	\$233.00	\$184.00	—	\$258.00	\$294.00	\$153.00
1998	Apr.	\$253.00	\$134.00	\$195.00	\$193.00	\$187.00	—	\$253.00	\$264.00	\$163.00
2000	Apr.	\$227.00	\$131.00	\$200.00	\$194.00	\$167.00	—	\$233.00	\$240.00	\$165.00
2002	Apr.	\$250.00	\$127.00	\$191.00	\$195.00	\$187.00	—	\$221.00	\$227.00	\$164.00
2004	Apr.	\$379.00	\$178.00	\$276.00	\$263.00	\$205.00	—	\$266.00	\$276.00	\$181.00
2006	Apr.	\$521.00	\$232.00	\$362.00	\$366.00	\$266.00	—	\$324.00	\$337.00	\$273.00
2008	Apr.	\$755.00	\$401.00	\$552.00	\$509.00	\$391.00	—	\$800.00	\$850.00	\$561.00
2010	Mar.	\$499.00	\$283.00	\$448.00	\$398.00	\$326.00	—	\$507.00	\$508.00	\$511.00
2012	Mar.	\$785.00	\$403.00	\$644.00	\$574.00	\$503.00	—	\$729.00	\$675.00	\$665.00
2014	Mar.	\$851.00	\$359.00	\$571.00	\$560.00	\$533.00	—	\$621.00	\$611.00	\$601.00

Current price reference (2024–2026, USDA AMS Illinois Production Cost Report)

The NASS price-paid series above ended in 2014. Current state-level fertilizer prices are published bi-weekly by USDA Agricultural Marketing Service. As of mid-May 2026 in Illinois:

Product	Price range (\$/ton)	Average	Trend
Anhydrous ammonia	\$995 – \$1,250	\$1,133	↑
Urea (46-0-0)	\$875 – \$925	\$902	→
Liquid Nitrogen (28-0-0)	\$500 – \$600	\$561	↑
Liquid Nitrogen (32-0-0)	\$650	\$650	↑
DAP (18-46-0)	\$840 – \$925	\$870	→
MAP (11-52-0)	\$840 – \$920	\$889	↓
Potash (white 0-0-62)	\$445 – \$560	\$504	→

Source: USDA AMS Illinois Production Cost Report (ams_3195.pdf), bi-weekly. Equivalent reports available for Iowa (ams_2863), Pennsylvania (ams_3726), and other states. These reports are the primary source for real-time benchmarking inside MarginScout.

Section 5 — Corn fertilizer application rates by state

Application rates in pounds per fertilized acre, from USDA NASS Agricultural Chemical Usage surveys. Most recent reported year is shown for each state along with the 2010s decade average. Used to validate whether a farmer's application program is in line with regional norms.

State	N latest yr	N lb/ac	N 2010s avg	P2O5 latest yr	P2O5 lb/ac	P2O5 2010s avg	K2O latest yr	K2O lb/ac	K2O 2010s avg
Illinois	2018	172	168	2018	108	96	2018	135	116
Indiana	2018	157	164	2018	64	70	2018	106	112
Iowa	2018	139	143	2018	74	70	2018	89	87
Kansas	2018	147	140	2018	60	44	2018	55	41
Kentucky	2018	148	158	2018	89	87	2018	108	100
Michigan	2018	144	131	2018	47	41	2018	96	91
Minnesota	2018	137	136	2018	74	64	2018	91	80
Missouri	2018	169	159	2018	71	68	2018	79	74
Nebraska	2018	165	149	2018	52	45	2018	32	28
North Dakota	2018	146	142	2018	62	56	2018	34	32
Ohio	2018	175	160	2018	73	71	2018	108	101
South Dakota	2018	126	127	2018	60	52	2018	55	42
Wisconsin	2018	113	105	2018	37	38	2018	76	64
U.S. average	2018	149	144	2018	69	64	2018	87	82

Reading the table: For Iowa corn, the most recent reported nitrogen application rate was 152 lb/ac in 2018, with a 2010s decade average around the same level. A farmer applying 200+ lb/ac is well above the regional norm and should question whether the recommendation is agronomically justified or driven by retailer incentives.

Section 6 — Soybean fertilizer application rates by state

Soybeans fix their own nitrogen, so N application is usually low. Phosphate and potash are the primary fertilizer inputs. The percentage of soybean acreage that receives nitrogen fertilizer is also a useful sanity check — in most Midwest states, fewer than 30% of soybean acres receive any nitrogen at all.

State	N % acres	N lb/ac	P2O5 % acres	P2O5 lb/ac	K2O % acres	K2O lb/ac
Arkansas	7	35	51	68	56	85
Illinois	25	22	43	75	61	111
Indiana	28	19	33	56	50	105
Iowa	13	14	21	59	32	86
Kansas	29	17	38	43	18	57
Kentucky	32	33	74	74	85	109
Michigan	50	15	48	42	77	88
Minnesota	29	14	35	46	36	68
Mississippi	6	21	31	63	44	103
Missouri	11	18	33	46	44	75
Nebraska	36	14	43	62	18	51
North Dakota	55	10	62	34	18	16
Ohio	33	14	37	59	57	103
South Dakota	42	12	55	46	28	34
Wisconsin	37	20	50	63	80	106
U.S. average	29	17	42	55	43	87

Reading the table: U.S. average soybean acreage receiving nitrogen was about 17% in 2018, at a rate of 17 lb/ac on those fields that did get N. Phosphate and potash are applied to roughly 30-40% of soybean acres at typical rates of 50-90 lb/ac. A farmer being pitched 150 lb/ac of nitrogen for soybeans should ask hard questions.

Section 7 — Source attribution and methodology

Primary data sources

USDA Economic Research Service (ERS) — Commodity Costs and Returns. Annual cost-of-production estimates for U.S. field crops at the national and ERS Farm Resource Region level. Methodology follows the American Agricultural Economics Association Task Force standards. Underlying data comes from the Agricultural Resource Management Survey (ARMS) on a 4-8 year rotation. Survey base years for corn are 1996, 2001, 2005, 2010, 2016, and 2021. Estimates between base years are updated annually using NASS price, acreage, and production data. Last updated May 2026.

USDA National Agricultural Statistics Service (NASS) — Agricultural Chemical Usage. State-level data on percentage of acreage receiving each nutrient and application rates per fertilized acre. Coverage is by selected states and selected years; not all state-crop-nutrient combinations are surveyed in every cycle. Most recent reporting cycle for corn and soybeans was 2018.

USDA NASS — Agricultural Prices (price paid series). National farm-level fertilizer prices, collected through 2014 when the series was discontinued. For current price benchmarking, refer to USDA Agricultural Marketing Service (AMS) state-level Production Cost Reports, which publish bi-weekly fertilizer and fuel prices.

ERS Farm Resource Regions (geographic definitions)

Region	States included
Heartland	Iowa, Illinois, Indiana, Ohio, Missouri (central Corn Belt)
Northern Crescent	New York, Pennsylvania, Michigan, Wisconsin, Minnesota
Northern Great Plains	Montana, North Dakota, South Dakota, Wyoming, parts of Nebraska
Prairie Gateway	Kansas, Oklahoma, Texas, parts of Colorado and New Mexico
Eastern Uplands	Kentucky, Tennessee, West Virginia, parts of Virginia and North Carolina
Southern Seaboard	Coastal Carolinas, Georgia, Alabama, parts of Florida
U.S. total	National weighted average across all regions

Intended use within MarginScout

This knowledge base is the seed dataset for the MarginScout benchmarking engine. Specifically:

Section 1 (2025 regional benchmarks) feeds the peer-comparison view. When a farmer submits costs, their line items are compared against the region matching their county.

Section 2 (10-year Heartland trends) feeds the "is this normal?" tooltip and the year-over-year change calculations.

Section 3 (30-year national history) feeds the downside-scenario calibration: the historical price and yield ranges define defensible bounds for what-if modeling.

Section 4 (fertilizer prices) feeds the real-time quote-verdict engine. Combined with the live USDA AMS bi-weekly state reports, it provides both historical context and current benchmarks for each parsed line item.

Sections 5 and 6 (application rates) feed the sanity-check layer. When a fertilizer quote arrives, the system checks not only the unit price but also whether the recommended application rate (lb/ac of N, P, K) is in line with the regional norm for that crop.

End of knowledge base document.

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